

BASE ARTURO PRAT, Antarctica

WMO: 890570

Lat: 62.479S

Lon: 59.665W

Elev: 16

StdP: 14.69

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.5	5.0	-3.2	4.6	0.9	1.4	5.9	5.9	59.3	4.0	54.7	7.4	23.6	80	1.034

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.1	40.0	37.5	39.0	36.9	38.3	36.4	38.5	39.2	37.7	38.3	37.1	37.7	10.1	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
37.7	33.2	38.4	37.0	32.3	37.8	36.3	31.5	37.3	14.7	39.4	14.3	38.5	14.1	37.9	44.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 47.2	40.7	34.1	DB	-0.1	45.7	6.2	4.1	-4.5	48.7	-8.2	51.1	-11.6	53.4	-16.1	56.4	(4)
(5)			WB	-0.5	40.8	5.8	2.1	-4.7	42.2	-8.1	43.5	-11.4	44.6	-15.6	46.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	28.6	35.2	35.2	33.4	30.0	27.6	23.6	21.4	22.3	23.5	27.6	30.3	33.0	(6)
(7)		DBStd	7.09	1.95	2.53	3.30	5.11	5.17	6.69	7.93	7.45	7.27	4.56	2.91	2.21	(7)
(8)		HDD50	7828	459	414	515	599	695	793	886	859	794	696	591	527	(8)
(9)		HDD65	13303	924	834	980	1049	1160	1243	1351	1324	1244	1161	1041	992	(9)
(10)		CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(10)
(11)		CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)
(13)		CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	13.6	10.6	11.2	12.5	14.6	14.1	16.1	15.1	15.2	14.7	14.9	12.2	11.8	(14)
(15)	Precipitation	PrecAvg	40.9	3.2	3.2	4.7	4.0	3.8	3.0	2.8	3.0	3.2	3.9	3.3	2.8	(15)
(16)		PrecMax	54.8	4.5	4.9	7.4	6.0	6.3	4.5	4.3	5.6	5.4	6.7	4.5	4.5	(16)
(17)		PrecMin	33.7	1.9	1.9	2.8	2.3	1.7	1.6	1.3	0.6	1.1	2.0	1.2	1.2	(17)
(18)		PrecStd	4.6	0.6	0.7	1.1	0.9	1.1	0.7	0.9	1.1	0.9	1.0	0.8	0.8	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	42.2	41.5	38.3	38.3	35.0	34.1	34.0	34.8	35.7	37.5	39.5	(19)
(20)			MCWB	38.2	38.6	39.3	36.5	35.2	33.6	32.6	32.6	33.3	33.5	34.5	35.9	(20)
(21)		2%	DB	39.7	39.9	39.1	37.0	35.6	33.7	32.9	32.8	33.3	34.2	35.9	37.7	(21)
(22)			MCWB	37.1	37.5	37.2	35.6	34.4	32.6	32.0	31.8	32.2	32.8	34.1	35.1	(22)
(23)		5%	DB	38.7	39.0	38.2	36.3	34.7	32.8	32.1	32.0	32.6	33.4	34.8	36.8	(23)
(24)			MCWB	36.5	37.1	36.7	35.1	33.7	31.9	31.4	31.4	31.8	32.4	33.5	34.6	(24)
(25)		10%	DB	37.9	38.3	37.4	35.5	33.8	32.0	31.2	31.1	32.0	32.7	34.0	35.9	(25)
(26)			MCWB	36.0	36.6	36.2	34.5	33.0	31.2	30.5	30.5	31.4	31.8	32.8	34.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.5	39.6	39.9	37.2	35.9	34.0	33.3	33.2	33.8	34.3	35.9	37.5	(27)
(28)			MADB	40.3	40.7	40.7	37.4	36.8	34.4	33.4	33.4	34.2	34.7	36.6	38.2	(28)
(29)		2%	WB	38.1	38.5	38.0	36.3	34.8	33.0	32.4	32.3	32.7	33.3	34.8	36.1	(29)
(30)			MADB	38.8	39.2	38.6	36.6	35.3	33.4	32.6	32.6	33.0	33.7	35.3	36.8	(30)
(31)		5%	WB	37.4	37.8	37.3	35.5	34.0	32.2	31.6	31.5	32.1	32.7	34.0	35.4	(31)
(32)			MADB	38.0	38.4	37.8	36.0	34.4	32.6	31.9	31.9	32.4	33.2	34.5	36.0	(32)
(33)		10%	WB	36.7	37.2	36.6	34.8	33.2	31.4	30.6	30.7	31.4	32.1	33.2	34.7	(33)
(34)			MADB	37.4	37.9	37.2	35.4	33.7	31.9	31.1	31.1	31.8	32.5	33.7	35.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.1	4.4	4.9	5.8	6.7	7.7	7.3	7.4	5.3	4.3	4.1	(35)
(36)			MCDBR	5.2	5.0	4.7	4.6	5.5	5.6	7.0	5.8	5.5	5.1	5.0	5.4	(36)
(37)			MCWBR	4.0	4.2	4.2	4.8	5.4	5.7	7.1	6.2	5.4	4.7	4.3	4.3	(37)
(38)		5% WB	MCDBR	4.5	4.3	4.2	4.4	4.8	5.6	6.8	5.6	5.3	4.7	4.4	4.6	(38)
(39)			MCWBR	3.9	4.0	4.1	4.7	5.1	5.8	7.0	6.0	5.3	4.6	4.1	4.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.308	0.308	0.297	0.280	0.223	0.366	0.208	0.268	0.289	0.310	0.304	0.309	(40)	
(41)		taud	2.499	2.508	2.507	2.374	2.086	2.814	2.035	2.233	2.364	2.403	2.463	2.474	(41)	
(42)		Ebn at Noon	306	291	266	213	136	91	135	207	261	286	306	310	(42)	
(43)		Edh at Noon	31	27	23	16	10	6	10	18	25	30	32	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1650	1189	626	274	77	32	56	201	537	1005	1565	1860	(44)	
(45)		RadStd	145	82	56	21	7	1	3	15	42	67	92	149	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)			N/A	N/A	N/A	N/A	-0.47	-0.76	N/A	N/A	N/A

Nomenclature: See separate page